Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Chemistry *study guide*

Chemistry explains what everything is made of, how substances react and how raw materials become medicines, fuels, fertilisers and food.

Big ideas

Everything is made of particles

Atoms make up elements and compounds. The numbers of protons, neutrons and electrons in an atom explain the order of the Periodic Table.

Bonding decides properties

Ionic, covalent and metallic bonds explain why salt melts at 801 °C while water boils at 100 °C, and why metals conduct electricity.

Reactions can be measured

The mole lets chemists count particles by weighing them, so they can predict exactly how much product a reaction will make.

Topic checklist

tick each topic when you can explain it

O LEVEL

- [ ] The particulate nature of matter
- [ ] Experimental techniques and separation
- [ ] Atomic structure and the Periodic Table
- [ ] Chemical bonding
- [ ] Formulae, equations and the mole
- [ ] Electrolysis
- [ ] Energy changes in reactions
- [ ] Rates of reaction
- [ ] Reversible reactions and equilibrium
- [ ] Acids, bases and salts
- [ ] Metals and the reactivity series
- [ ] Organic chemistry

A LEVEL

- [ ] Atoms, molecules and stoichiometry
- [ ] Electron configuration
- [ ] Chemical bonding and intermolecular forces
- [ ] States of matter
- [ ] Chemical energetics
- [ ] Electrochemistry
- [ ] Equilibria
- [ ] Reaction kinetics
- [ ] Periodicity, Group 2 and Group 17
- [ ] Organic chemistry and analysis

Key terms

Ion	An atom or group of atoms with an electrical charge, formed by losing or gaining electrons.
Isotopes	Atoms of the same element with different numbers of neutrons.
Covalent bond	A shared pair of electrons between two non-metal atoms.
Ionic bond	The strong attraction between oppositely charged ions, formed when electrons are transferred.
Mole	The amount of substance that contains 6.02×10^{23} particles.
Catalyst	A substance that speeds up a reaction without being used up.
Oxidation	The loss of electrons, or the gain of oxygen, by a substance.
Electrolysis	Breaking down an ionic compound, molten or in solution, using electricity.

In the laboratory

- Filtration, evaporation and crystallisation
- Paper chromatography
- Titration with a burette and pipette
- Gas tests: hydrogen ‘squeaky pop’, oxygen relights a glowing splint, carbon dioxide turns limewater milky
- Measuring the rate of a reaction
- Electrolysis of solutions

Study smarter

1. Learn the first 20 elements and the common ions.
2. Balance every equation and add state symbols.
3. Set out mole calculations step by step, with units.
4. Learn each test together with its observation.

Check yourself

1. A chlorine atom has the electron arrangement 2,8,7. What ion does it form?

2. Which gas relights a glowing splint?

3. Balance the equation: $H_2 + O_2 \rightarrow H_2O$

Answers 1. Cl^-. It gains one electron to complete its outer shell (2,8,8). · 2. Oxygen. · 3. $2H_2 + O_2 \rightarrow 2H_2O$

Where it leads: Pharmacy, Chemical engineering, Mining and metallurgy, Food science, Medicine, Forensic science